MECH 151: Technical Design – Prototyping

# Semester Design Portfolio

*Template*

Leeward Community College · Mechatronics Program

Fall 2026

**Student Name:** _____

**Instructor:** Kainani "Kai" Santos

**Semester:** Fall 2026 (Aug 24 – Dec 18, 2026)

# Submitting Your Digital CAD Files

These instructions apply to every unit's files. Your completed portfolio — all three units together — is submitted directly to the instructor by email, and is worth 50% of your course grade per the syllabus grading policy.

## File Formats to Include

- Native file(s) in your CAD software's format (e.g., .sldprt, .sldasm, .slddrw, or your software's equivalent) so your full parametric feature tree is preserved and can be graded.
- A neutral exchange copy of each part and assembly in STEP format (.step or .stp), consistent with the CAD/CAM data translation practices covered in Unit 1.
- A PDF of every drawing sheet (multiview, detail, and assembly drawings), in addition to the native drawing file.

## File Naming

- Use one consistent naming pattern for every file: LastName_FirstName_UnitX_ShortPartName.ext — e.g., Santos_Kai_Unit1_BracketPart.sldprt.
- Keep names short and use underscores instead of spaces or special characters.

## Folder Structure

- Organize your submission into one folder per unit — Unit1_CADFundamentals, Unit2_DFA, Unit3_DFM — each containing that unit's native files, STEP exports, and drawing PDFs.
- Keep an assembly and every part file it references together in the same folder so the assembly opens without missing references.

## How and When to Submit

- Compress your complete portfolio folder (all three units) into a single .zip file before uploading.
- Email the .zip to kainanis@hawaii.edu as your Design Portfolio submission — either as a direct attachment, or through the UH FileDrop service (hawaii.edu/filedrop, recommended since a full portfolio .zip is usually too large for email) addressed to the same email — by the due date announced in class and on the syllabus.
- Before submitting, confirm your .zip opens cleanly and every file is present — corrupted or incomplete uploads cannot be graded.

# How to Use This Portfolio

This template tracks your design work across all three units of MECH 151. Each unit follows the same three-part structure:

- What's Evaluated — the specific skills and deliverables graded for that unit, drawn directly from the unit topics.
- Your Design — an open-ended section where you present a design of your own that demonstrates proficiency in that unit's topics. This is not fill-in-the-blank: describe your own problem, approach, and reasoning, and support it with screenshots, drawings, or photos of your work.
- Designer Feedback — a running log where you record feedback you received (from the instructor, a peer review, or your own testing) and how you plan to address it, consistent with the cyclic design process (define, design, build, test, evaluate, and loop back).

Duplicate this file (or copy its sections into your own document) and complete each unit as you finish the corresponding portfolio module. The evaluation overview below summarizes what each unit is graded on before you get to the detailed, unit-by-unit sections.

**The answer boxes in this template are sized to keep the document compact, not to cap what you write. Include all information pertinent to your evaluation — expand a box as far as you need to fully describe your design, approach, and reasoning.**

# Portfolio Evaluation Overview

The table below states, unit by unit, exactly what your portfolio entry is evaluated on. Detailed criteria and quiz/due dates for each unit are repeated at the start of that unit's section.

| Unit | What's Being Evaluated (Portfolio Focus) | Evaluated / Due |
|---|---|---|
| **Unit 1 — CAD Fundamentals** | Controlled sketches, a multiview drawing (with section and auxiliary views as needed), and a first fully parametric part model with correctly configured templates. | Unit 1 Quiz — Week 8 |
| **Unit 2 — Design for Assembly (DFA)** | Correctly called-out thread and fastener notes, a multi-part assembly model with an exploded assembly drawing and BOM, plus a documented design revision. | Unit 2 Quiz — Week 11 |
| **Unit 3 — Design for Manufacturing (DFM)** | Correctly stated tolerances and fits, manufacturing-ready files, and GD&T documentation demonstrating your design-for-manufacturing reasoning. | Unit 3 Quiz — Week 16 |

*Across every unit, your work is also judged against the Design Fundamentals principles that precede Unit 1: clearly defined requirements/objectives/constraints, deliberate design intent, fully defined and robust models, and evidence that you evaluated your design rather than stopping at a first pass.*

# Unit 1 — CAD Fundamentals

## Sketching, Views, CAD Configuration, and the First Parametric Model

*Build the drawing and modeling foundation every later unit depends on — how to lay out a controlled sketch, how to project a part into standard and specialized views, how to configure your CAD environment, and how to turn that sketch into a parametric solid model.*

### What's Evaluated in This Unit

- Sketches are fully constrained/defined and use correct 2D construction geometry (tangency, fillets/chamfers, bisectors, offsets, symmetry).
- Multiview drawing correctly projects primary and secondary views with true length/shape relationships and follows accepted dimensioning conventions.
- Section and auxiliary views are chosen and constructed correctly — full/half/offset sections where appropriate, and true-shape auxiliary views for inclined surfaces.
- CAD software is configured consistently (templates, document settings, custom properties), and models translate cleanly to neutral exchange formats (e.g., STEP) when required.
- Parametric model captures clear design intent — deliberate feature order, stable references, features that edit and rebuild cleanly.
- Relevant measurable properties (area, centroid, moments of inertia, volume, mass, center of gravity) are understood and correctly applied where the design calls for them.
- No single "right" feature tree is expected — the model is judged on captured intent, clean rebuilds, and editability.

**Assessment: Unit 1 Quiz — Week 8**

### Your Design — Demonstrating Proficiency

Present an open-ended design of your own choosing that shows you can apply this unit's topics. Describe the problem or goal, your approach, and the reasoning behind your decisions — then support it with your model, drawing, or file evidence.

#### Design Overview

*Describe the part or assembly, the problem/requirement it addresses, and any objectives or constraints you set for it.*

#### Design Approach & Intent

*Explain the key decisions behind your design — sketch/feature strategy, views chosen, tolerances, joining method, or manufacturing process, as applicable to this unit — and why.*

#### Supporting Evidence

*[Insert CAD screenshots, drawings, or photos demonstrating this unit's work here]*

### Designer Feedback — Addressing Future Work

Log feedback you received (instructor, peer review, self-testing) and your plan to address it. Carry unresolved items forward into later units when relevant.

| Feedback / Issue Identified | Plan to Address / Future Work | Status |
|---|---|---|
| | | |

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|---|---|---|
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*Add more rows to this table as needed — this is a starting point, not a limit.*

# Unit 2 — Design for Assembly (DFA)

## Threads, Fasteners, and Documenting Assemblies

*Represent threads and fasteners with correct callouts, then combine part models into working assemblies, document them the way a manufacturer expects, and apply the design principles that make an assembly easier and cheaper to put together.*

### What's Evaluated in This Unit

- Threads and fasteners are represented and called out correctly (thread notes, class of fit, clearance/tap-drill holes sized appropriately).
- Assembly is built with correct mates/constraints; components are positioned, aligned, and managed cleanly in the assembly tree.
- Joining method(s) — fasteners, inserts, snap-fits, press fits, compliant/living hinges, or permanent joints — are chosen deliberately, with DFA trade-offs considered.
- Catalog/imported hardware (e.g., McMaster-Carr, TraceParts, GrabCAD) is brought in cleanly and carries a real part number in the BOM.
- Exploded view, balloons, and bill of materials are complete and easy to read.
- DFA principles are applied — reduced part count, ease of assembly, tolerance stack-up considered.
- A design review is documented: what was evaluated, what feedback was received, and how the assembly was revised in response.

**Assessment: Unit 2 Quiz — Week 11**

### Your Design — Demonstrating Proficiency

Present an open-ended design of your own choosing that shows you can apply this unit's topics. Describe the problem or goal, your approach, and the reasoning behind your decisions — then support it with your model, drawing, or file evidence.

#### Design Overview

*Describe the part or assembly, the problem/requirement it addresses, and any objectives or constraints you set for it.*

#### Design Approach & Intent

*Explain the key decisions behind your design — sketch/feature strategy, views chosen, tolerances, joining method, or manufacturing process, as applicable to this unit — and why.*

#### Supporting Evidence

*[Insert CAD screenshots, drawings, or photos demonstrating this unit's work here]*

### Designer Feedback — Addressing Future Work

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# Unit 3 — Design for Manufacturing (DFM)

## Tolerancing and From Model to Prototype

*State tolerances and fits appropriately, then close the loop from CAD model to physical part — preparing manufacturing-ready files, understanding what each process can and cannot make, and operating shop equipment safely to produce a prototype.*

### What's Evaluated in This Unit

- Tolerances and fits (limit dimensions, plus/minus tolerances, fit classes) are applied appropriately and stated unambiguously.
- GD&T callouts (datums, feature control frames, form/orientation/location/profile/runout symbols, material condition modifiers) are correct and unambiguous.
- Files are manufacturing-ready for the chosen process — correct export format, toolpaths/checks completed as applicable.
- Design decisions suit the selected process and material (additive, subtractive/CNC, or laser) — DFM considerations are addressed, not just described.
- Manufacturing-ready files are checked for completeness and consistency before hand-off — correct units, no missing tolerances or callouts, geometry consistent with the intended process.

**Assessment: Unit 3 Quiz — Week 16**

### Your Design — Demonstrating Proficiency

Present an open-ended design of your own choosing that shows you can apply this unit's topics. Describe the problem or goal, your approach, and the reasoning behind your decisions — then support it with your model, drawing, or file evidence.

#### Design Overview

*Describe the part or assembly, the problem/requirement it addresses, and any objectives or constraints you set for it.*

#### Design Approach & Intent

*Explain the key decisions behind your design — sketch/feature strategy, views chosen, tolerances, joining method, or manufacturing process, as applicable to this unit — and why.*

#### Supporting Evidence

*[Insert CAD screenshots, drawings, or photos demonstrating this unit's work here]*

### Designer Feedback — Addressing Future Work

Log feedback you received (instructor, peer review, self-testing) and your plan to address it. Carry unresolved items forward into later units when relevant.

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